

Table S7. Geographic and photoperiod associations

	Overall site FDR- adjusted p	Photoperiod		Latitude		Site-versus- latitude FDR- adjusted p	Exact fitted sample
		Photoperiod association (95% CI)	Photoperiod FDR- adjusted p	Latitude association per 10° (95% CI)	Latitude FDR- adjusted p		
Dynamics							
Interdaily stability	0.228	×0.98 [0.94– 1.02]	0.401	×1.01 [0.96– 1.06]	0.903	0.175	$n_{\text{participants}} = 141$; $n_{\text{participant-days}} = 816$
Intradaily variability	0.483	-0.02 [-0.06– 0.02]	0.327	-0.04 [-0.08– 0.01]	0.139	0.723	$n_{\text{participants}} = 141$; $n_{\text{participant-days}} = 816$
Level							
Mean melEDI	<0.001	×1.18 [1.11– 1.26]	<0.001	×1.15 [1.07– 1.25]	0.002	0.032	$n_{\text{participants}} = 141$; $n_{\text{participant-days}} = 816$
Brightest 10 h mean	0.011	×1.24 [1.13– 1.36]	<0.001	×1.24 [1.11– 1.39]	0.001	0.325	$n_{\text{participants}} = 141$; $n_{\text{participant-days}} = 816$
Darkest 10 h mean	<0.001	×1.09 [1.03– 1.14]	0.002	×1.04 [0.97– 1.11]	0.400	<0.001	$n_{\text{participants}} = 141$; $n_{\text{participant-days}} = 816$
Duration							
Time above 1,000 lx melEDI	0.071	×1.21 [1.13– 1.29]	<0.001	×1.03 [0.94– 1.12]	0.769	0.053	$n_{\text{participants}} = 141$; $n_{\text{participant-days}} = 816$
Time above 250 lx melEDI during wake	0.007	×1.14 [1.07– 1.20]	<0.001	×1.12 [1.04– 1.20]	0.013	0.053	$n_{\text{participants}} = 141$; $n_{\text{participant-days}} = 737$
Time below 10 lx melEDI before sleep	0.049	-0.13 [-0.20 –0.07] h	<0.001	0.00 [-0.09–0.09] h	0.962	0.040	$n_{\text{participants}} = 139$; $n_{\text{participant-days}} = 655$
Time below 1 lx melEDI during sleep	0.058	×0.99 [0.97– 1.01]	0.227	×1.00 [0.97– 1.02]	0.903	0.043	$n_{\text{participants}} = 141$; $n_{\text{participant-days}} = 778$
Longest continuous period above 250 lx melEDI	0.398	×1.12 [1.07– 1.19]	<0.001	×1.06 [0.99– 1.13]	0.139	0.626	$n_{\text{participants}} = 141$; $n_{\text{participant-days}} = 816$

	Overall site FDR- adjusted p	Photoperiod		Latitude		Site-versus- latitude FDR- adjusted p	Exact fitted sample
		Photoperiod association (95% CI)	Photoperiod FDR- adjusted p	Latitude association per 10° (95% CI)	Latitude FDR- adjusted p		
Timing							
Midpoint of the brightest 10 hours	<0.001	0.05 [-0.05–0.15] h	0.305	0.19 [0.06–0.32] h	0.013	0.010	$n_{\text{participants}} = 141$; $n_{\text{participant-days}} = 816$
Midpoint of the darkest 10 hours	0.020	-0.15 [-0.25 –0.05] h	0.004	0.12 [-0.00–0.24] h	0.114	0.043	$n_{\text{participants}} = 141$; $n_{\text{participant-days}} = 816$
Mean timing of exposure above 250 lx melEDI	<0.001	0.12 [0.03–0.21] h	0.013	0.33 [0.21–0.46] h	<0.001	<0.001	$n_{\text{participants}} = 141$; $n_{\text{participant-days}} = 742$
First light timing above 250 lx melEDI	0.112	-0.10 [-0.25– 0.05] h	0.227	0.02 [-0.17–0.21] h	0.903	0.077	$n_{\text{participants}} = 140$; $n_{\text{participant-days}} = 727$
Last light timing above 250 lx melEDI	<0.001	0.32 [0.19–0.46] h	<0.001	0.46 [0.27–0.64] h	<0.001	<0.001	$n_{\text{participants}} = 141$; $n_{\text{participant-days}} = 687$
Exposure history							
melEDI dose	0.209	×1.28 [1.17– 1.39]	<0.001	×1.02 [0.91– 1.13]	0.903	0.156	$n_{\text{participants}} = 141$; $n_{\text{participant-days}} = 761$
Spectrum							
Melanopic daylight efficacy ratio	<0.001	0.02 [0.02–0.03]	<0.001	-0.01 [-0.02 –0.00]	0.013	0.010	$n_{\text{participants}} = 137$; $n_{\text{participant-days}} = 702$

This summary combines the model-level decisions, photoperiod and latitude associations, 95% CIs, and exact fitted samples. Each displayed p-value is FDR-adjusted within its explicitly labelled complete 17-test family; bold values meet the FDR-adjusted $p < 0.050$ rule. Full raw p-values and test statistics remain in the detailed tables below. Exact fitted samples use italic *n* with participant and participant-day subscripts. For the 15 participant-day models, the participant-day count equals the number of fitted observations. The two participant-level dynamics models use one fitted observation per participant and retain the participant-day count as contributing repeated-day support. All models include nine sites.

Supplementary Table S7. Overall-site, civil-photoperiod and absolute-latitude associations for the 17 near-eye personal light-exposure metrics. Each displayed *p* value is FDR-adjusted within its stated 17-test family. Site and latitude were evaluated in separate models because absolute latitude is fixed within and confounded with study site.